

Tourist Reservation System using C++

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Source Code



Hello, coders. Welcome to the Coding_knowledge Documet. In this Document, we will learn how to create a Tourist Reservation System using C++.

The Tourist Reservation System is built on the idea of keeping track of excursions and tourist reservations. The entire system is written in C++ and is a simple console application.

The system includes both the **admin** and **user** sides.

- The admin can do everything here, including adding, examining, and changing trip data and reservations.
- The user can make a reservation, cancel the reservation and search for a reservation.

This is a basic Tourist reservation management system which makes use of file management and displays the programming language's class and object. In the system, the user can only perform simple tasks such as adding a new destination, cancelling a reservation, and seeing available tours. The system includes all of the required functions for booking a tour spot. You only need to input the necessary information to utilize this system. This project will show you how to use the C++ programming language to add, list, alter, and remove records in a file.

Features of the Program:

- **Create new Trip reservations**
- **Display a Trip**
- **Display a Reservation**
- **Update Trip**
- **Confirm Status**
- **Cancel a Reservation**

Tourist Reservation System using C++ Source code:

You can directly use this code by copying it in your IDE to understand the working and then can create it by understanding the project.

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```
#include<iostream>
```

```
#include<fstream>
```

```
#include<string.h>
```

```
using namespace std;
```

```
char adminf[20]="Trip.txt";
```

```
char userf[20]="Reservation.txt";
```

```
struct tour{
```

```
    char
```

```
    ad_place[20],ad_code[20],ad_country[20],ad_days[20],a  
    d_prices[20];
```

```
    char
```

```
    us_phone[20],us_name[20],us_people[20],us_date[20],  
    us_status[20],us_dcode[20];
```

```
};
```

```
class Tourist{
```

public:

void ad_pack(tour r);

tour ad_unpack(char a[]);

void ad_writedata();

void ad_display();

void ad_modify();

void us_pack(tour r);

tour us_unpack(char a[]);

void us_writedata();

void us_display();

void us_search();

void us_modify();

};

// ADMIN PANEL

void Tourist ::ad_pack(tour r){

fstream fp;

fp.open(adminf,ios::out|ios::app);

if(!fp){

```
    cout<<"\nCannot open file";  
    exit(0);  
}  
  
char buff[45];  
  
strcpy(buff,r.ad_code);  
strcat(buff,"|");  
strcat(buff,r.ad_place);  
strcat(buff,"|");  
strcat(buff,r.ad_country);  
strcat(buff,"|");  
strcat(buff,r.ad_days);  
strcat(buff,"|");  
strcat(buff,r.ad_prices);  
strcat(buff,"|");  
for(int i=0;i<45-strlen(buff);i++)  
    strcat(buff,"|");  
fp<<buff<<endl;  
fp.close();  
}
```

```
tour Tourist::ad_unpack(char buff[]){  
    tour r;  
    int i=0,j=0;  
    while(buff[j]!='|')  
        r.ad_code[i++]=buff[j++];  
    r.ad_code[i]='\0';  
    i=0;  
    j++;  
    while(buff[j]!='|')  
        r.ad_place[i++]=buff[j++];  
    r.ad_place[i]='\0';  
    i=0;  
    j++;  
    while(buff[j]!='|')  
        r.ad_country[i++]=buff[j++];  
    r.ad_country[i]='\0';  
    i=0;  
    j++;
```

```
while(buff[j]!='|')  
    r.ad_days[i++]=buff[j++];  
r.ad_days[i]='\0';  
i=0;  
j++;  
while(buff[j]!='|')  
    r.ad_prices[i++]=buff[j++];  
r.ad_prices[i]='\0';  
return(r);  
}
```

```
void Tourist::ad_writedata(){  
    tour s;  
    cout<<"Enter the Destination Code\nD-";  
    cin>>s.ad_code;  
    cout<<"Enter the Destination Place\n";  
    cin>>s.ad_place;  
    cout<<"Enter the Country of the Destination\n";  
    cin>>s.ad_country;
```



```
cout<<"Enter the Number of days of the Trip\n";  
cin>>s.ad_days;  
cout<<"Enter the Cost of the trip\n";  
cin>>s.ad_prices;  
ad_pack(s);  
}
```

```
void Tourist::ad_display(){
```

```
    fstream fp;
```

```
    char buff[45];
```

```
    tour r;
```

```
    fp.open(adminf,ios::in);
```

```
    if(!fp){
```

```
        cout<<"\nCannot open file";
```

```
        exit(0);
```

```
    }
```

```
    cout<<"-----  
-----\n";
```

```
cout<<"Sr No.\t\tDEST.  
CODE\tPLACE\t\tCOUNTRY\t\tDAYS\t\tPRICES\n";
```

```
cout<<"-----  
-----\n";
```

```
int c=1;
```

```
while(1){
```

```
    fp.getline(buff,45);
```

```
    if(fp.eof())break;
```

```
        r=ad_unpack(buff);
```

```
        cout<<c++<<"\t\t"<<r.ad_code<<"\t\t"  
<<r.ad_place<<"\t\t"<<r.ad_country<<"\t\t"  
<<r.ad_days<<"\t\t"<<r.ad_prices<<endl;
```

```
    }
```

```
    fp.close();
```

```
    return;
```

```
}
```

```
void Tourist ::ad_modify(){
```

```
    fstream fp;
```

```
char ad_code[15],buff[45];
```

```
int i,j,ch;
```

```
tour s[100];
```

```
fp.open(adminf,ios::in);
```

```
if(!fp){
```

```
    cout<<"\nCannot open file";
```

```
    exit(0);
```

```
}
```

```
    cout<<"\nENTER THE DESTINATION CODE TO BE  
MODIFIED: ";
```

```
    cin>>ad_code;
```

```
    i=0;
```

```
    while(1){
```

```
        fp.getline(buff,45);
```

```
        if(fp.eof())break;
```

```
        s[i]=ad_unpack(buff);
```

```
        i++;
```

```
}
```

```
    for(j=0;j<i;j++){
```

```
if(strcmp(s[j].ad_code,ad_code)==0){  
    cout<<"VALUES OF THE TRIP\n";  
    cout<<"\nDestination Code: "<<s[j].ad_code;  
    cout<<"\nDestination Place: "  
<<s[j].ad_place;  
    cout<<"\nDestination Country: "  
<<s[j].ad_country;  
    cout<<"\nNo of days of Trip: "<<s[j].ad_days;  
    cout<<"\nTotal Cost of the Trip: "  
<<s[j].ad_prices;  
    cout<<"\nWhat you want to Update: ";  
    cout<<"\n\n\t\tEnter 1 for changing  
DESTINATION PLACE\n";  
    cout<<"\t\tEnter 2 for changing DAYS OF THE  
TRIP\n";  
    cout<<"\t\tEnter 3 for changing COST OF THE  
TRIP\n";  
    cout<<"\t\t\t";  
    cin>>ch;  
    switch(ch){
```

case 1:

cout<<"Destination Place: ";

cin>>s[j].ad_place;

cout<<"Destination Country: ";

cin>>s[j].ad_country;

break;

case 2:

cout<<"No of days of Trip: ";

cin>>s[j].ad_days;

break;

case 3:

cout<<"Total Cost of the Trip: ";

cin>>s[j].ad_prices;

break;

}

break;

}

}

if(j==i){

```
        cout<<"\n TRIP NOT FOUND";  
        return;  
    }  
    fp.close();  
    fstream fd;  
    fd.open(adminf,ios::out|ios::trunc);  
    if(!fd){  
        cout<<"\nFile Not Found";  
        exit(0);  
    }  
    for(j=0;j<i;j++)  
        ad_pack(s[j]);  
    fd.close();  
}
```

// USER PANEL

```
void Tourist ::us_pack(tour r){  
    fstream fp;
```

```
fp.open(userf,ios::out|ios::app);
if(!fp){
    cout<<"\nFile Not Found";
    exit(0);
}
char buff[45];
strcpy(buff,r.us_name);
strcat(buff,"|");
strcat(buff,r.us_phone);
strcat(buff,"|");
strcat(buff,r.us_people);
strcat(buff,"|");
strcat(buff,r.us_dcode);
strcat(buff,"|");
strcat(buff,r.us_date);
strcat(buff,"|");
strcat(buff,r.us_status);
strcat(buff,"|");
for(int i=0;i<45-strlen(buff);i++)
```

```
        strcat(buff,"|");
    fp<<buff<<endl;
    fp.close();
}

tour Tourist::us_unpack(char buff[]){
    tour r;
    int i=0,j=0;
    while(buff[j]!='|')
        r.us_name[i++]=buff[j++];
    r.us_name[i]='\0';
    i=0;
    j++;
    while(buff[j]!='|')
        r.us_phone[i++]=buff[j++];
    r.us_phone[i]='\0';
    i=0;
    j++;
```



```
while(buff[j]!='|')
    r.us_people[i++]=buff[j++];
r.us_people[i]='\0';
i=0;
j++;
while(buff[j]!='|')
    r.us_dcode[i++]=buff[j++];
r.us_dcode[i]='\0';
i=0;
j++;
while(buff[j]!='|')
    r.us_date[i++]=buff[j++];
r.us_date[i]='\0';
i=0;
j++;
while(buff[j]!='|')
    r.us_status[i++]=buff[j++];
r.us_status[i]='\0';
return(r);
```

```
}  
  
void Tourist::us_writedata(){  
    tour r;  
  
    cout<<"ENTER YOUR NAME: ";  
  
    cin>>r.us_name;  
  
    cout<<"ENTER YOUR CONTACT NO.: ";  
  
    cin>>r.us_phone;  
  
    cout<<"ENTER NO. OF TOURIST: ";  
  
    cin>>r.us_people;  
  
    cout<<"ENTER THE DESTINATION CODE: D-";  
  
    cin>>r.us_dcode;  
  
    cout<<"ENTER DATE OF JOURNEY (DD/MM/YY): ";  
  
    cin>>r.us_date;  
  
    cout<<"\nENTER 'Confirm' TO CONFIRM YOUR  
RESERVATION: ";  
  
    cin>>r.us_status;  
  
    us_pack(r);  
}
```

```
void Tourist::us_display(){
    fstream fp;
    char buff[45];
    tour r;
    fp.open(userf,ios::in);
    if(!fp){
        cout<<"\nCannot open file";
        exit(0);
    }

    cout<<"-----\n";
    -----\n";

    cout<<"Sr No.\tNAME\t\tCONTACT NO\t\tTOTAL
PEOPLE\t\tDATE\t\tDEST. CODE\t\tSTATUS\n";

    cout<<"-----\n";
    -----\n";

    int c=1;
    while(1){
        fp.getline(buff,45);
```

```
    if(fp.eof())break;

    r=us_unpack(buff);

    cout<<c++<<"\t"<<r.us_name<<"\t\t"
<<r.us_phone<<"\t\t"<<r.us_people<<"\t\t\t"
<<r.us_date<<"\t\t"<<r.us_dcode<<"\t\t"
<<r.us_status<<endl;

}

fp.close();

return;

}
```

```
void Tourist ::us_modify(){

    fstream fp;

    char us_name[15],buff[45];

    int i,j;

    tour s[100];

    fp.open(userf,ios::in);

    if(!fp){
```

```
cout<<"\nFile not Found";
```

```
exit(0);
```

```
}
```

```
cout<<"\nEnter Your name to cancel your  
reservation: ";
```

```
cin>>us_name;
```

```
i=0;
```

```
while(1){
```

```
fp.getline(buff,45);
```

```
if(fp.eof())break;
```

```
s[i]=us_unpack(buff);
```

```
i++;
```

```
}
```

```
for(j=0;j<i;j++){
```

```
if(strcmp(s[j].us_name,us_name)==0){
```

```
    cout<<"\nYour Reservation Details are:\n";
```

```
    cout<<"\nName: "<<s[j].us_name;
```

```
    cout<<"\nContact No.: "<<s[j].us_phone;
```

```
    cout<<"\nNo.of People: "<<s[j].us_people;
```

```
cout<<"\nDestination: "<<s[j].us_dcode;
cout<<"\nDate of Journey: "<<s[j].us_date;
cout<<"\nStatus: "<<s[j].us_status;

cout<<"\n\nENTER 'Cancel' TO CANCEL YOUR
TICKET: \n";

cin>>s[j].us_status;

cout<<"\nYour Reservation has been
Cancelled\n";

cout<<"\n\nName: "<<s[j].us_name;
cout<<"\nContact No.: "<<s[j].us_phone;
cout<<"\nNo.of People: "<<s[j].us_people;
cout<<"\nDestination: "<<s[j].us_dcode;
cout<<"\nDate of Journey: "<<s[j].us_date;
cout<<"\nStatus: "<<s[j].us_status;

break;
}
}

if(j==i){

cout<<"\nRECORD NOT FOUND";
```

```
        return;
    }

    fp.close();

    fstream fd;

    fd.open(userf,ios::out|ios::trunc);

    if(!fd){

        cout<<"\nCannot open file";

        exit(0);

    }

    for(j=0;j<i;j++)

        us_pack(s[j]);

    fd.close();

}
```

```
void Tourist::us_search(){

    fstream fp;

    char us_name[15],buff[45];

    int i,j;

    tour s;
```

```
fp.open(userf,ios::in);
if(!fp){
    cout<<"\nCannot open file";
    exit(0);
}
cout<<"\nENTER THE NAME TO BE SEARCHED: ";
cin>>us_name;
i=0;
while(1){
    fp.getline(buff,45);
    if(fp.eof())break;
    s=us_unpack(buff);
    if(strcmp(s.us_name,us_name)==0){
        cout<<"\n RESERVATION FOUND\n";
        cout<<"\nNAME: "<<s.us_name;
        cout<<"\nCONTACT NO.: "<<s.us_phone;
        cout<<"\nNO. OF PEOPLE: "<<s.us_people;
        cout<<"\nDATE OF JOURNEY: "<<s.us_date;
        cout<<"\nSTATUS: "<<s.us_status;
```



```

        return;
    }
}

cout<<"\nRESERVATION NOT FOUND";

return;
}

int main(){

    cout<<"                \t\t WELCOME TO TOURIST
RESERVATION SYSTEM\n\n\n";

    cout<<"                \t  TOURIST
RESERVATION SOFTWARE\n\n\n";

    cout<<"                \t    PRESS ENTER TO
CONTINUE....\n\n";

    if(cin.get()=='\n'){

        system("cls");

        int choice,ch1,ch2;

        Tourist obj;

        cout<<"\n Who are You?\n";

```

```
cout<<"1. ADMIN\n";
cout<<"2. USER\n\n";
cout<<"Enter any other key for exit\n\n\n";
cout<<"Enter Your Option\n";
cin>>choice;
switch(choice){
    case 1:
        system("cls");
        cout<<"WELCOME ADMIN\n";
        cout<<"Enter your option\n";
        cout<<"1. New trips\n";
        cout<<"2. Display a Trip\n";
        cout<<"3. Display a Reservation\n";
        cout<<"4. Update Trip\n";
        cout<<"Enter any other key for exit\n\n";
        while(1){
            cout<<"\nENTER UR CHOICE: ";
            cin>>ch1;
            switch(ch1){
```

case 1:

obj.ad_writedata();

break;

case 2:

obj.ad_display();

break;

case 3:

obj.us_display();

break;

case 4:

obj.ad_modify();

break;

default:

exit(0);

}

}

case 2:

system("cls");

cout<<"WELCOME USER\n";

```
cout<<"Enter your option\n";
cout<<"1. New Reservation\n";
cout<<"2. Search Reservation\n";
cout<<"3. Cancel Reservation\n";
cout<<"Enter any other key for exit\n\n\n";
while(1){
```

```
    cout<<"\nENTER UR CHOICE: ";
```

```
    cin>>ch2;
```

```
    switch(ch2){
```

```
        case 1:
```

```
            cout<<"-----";
```

```
            cout<<"\n Choose Your destination\n";
```

```
            cout<<"-----"
```

```
\n";
```

```
            obj.ad_display();
```

```
            cout<<"\n";
```

```
            obj.us_writedata();
```

```
            break;
```

```
        case 2:
```

```
        obj.us_search();  
        break;  
    case 3:  
        obj.us_modify();  
        break;  
    default:  
        exit(0);  
    }  
}  
  
default:  
    exit(0);  
}  
}  
}
```

Now let us understand the code:-

- After writing the header of the code with the required libraries, a structure is created – struct tour{}; for taking the

tour details such as place, code, country, days, prices, phone, name, number of people, date, status.

- We will create a class Tourist which contains all the functions required to perform the reservation task in this project.
- These are the functions declared:-void ad_pack(tour r), tour ad_unpack(char a[]), void ad_writedata(), void ad_display(), void ad_modify(), void us_pack(tour r), tour us_unpack(char a[]), void us_writedata(), void us_display(), void us_search(), void us_modify();
- Now outside the class we will define these functions using the scope resolution operator.
- The writedata() function takes in all the details –
- Destination Code, Destination Place, Country of the Destination,
- Number of days of the Trip, Cost of the trip.
- display() function will display all the data you entered in a tabular format.
- modify() function allows you to modify the values entered for the data for your tour reservation. It uses a while, for loop, if statement and switch case control statements.
- search() function allows you to search for a particular reservation from the database.
- Defining the main() function : It will display the heading for the project. It uses system("cls"); – Used to clear the screen. It will ask user : Who you are (admin or user). The user will select a choice and to call functions for that particular choice we will be using a switch case.

- This sums up our project of Tourist Reservation System using C++.

Final Output:-

Here is an example to show how this project works

```
WELCOME TO TOURIST RESERVATION SYSTEM
```

```
TOURIST RESERVATION SOFTWARE
```

```
PRESS ENTER TO CONTINUE....
```

```
Who are You?
```

```
1. ADMIN
```

```
2. USER
```

```
Enter any other key for exit
```

```
Enter Your Option
```



Upon selecting 1 : Admin

WELCOME ADMIN

Enter your option

1. New trips

2. Display a Trip

3. Display a Reservation

4. Update Trip

Enter any other key for exit

ENTER UR CHOICE:

Entering details as Admin


```
WELCOME ADMIN
Enter your option
1. New trips
2. Display a Trip
3. Display a Reservation
4. Update Trip
Enter any other key for exit

ENTER UR CHOICE: 1
Enter the Destination Code
D-mumbai
Enter the Destination Place
mumbai
Enter the Country of the Destination
india
Enter the Number of days of the Trip
7
Enter the Cost of the trip
50000

ENTER UR CHOICE: █
```

**Using display() function,
all details are displayed in
a tabular format.**

```
ENTER UR CHOICE: 2
```

Sr No.	DEST. CODE	PLACE	COUNTRY	DAYS	PRICES
1	mumbai	mumbai	india	7	50000

```
ENTER UR CHOICE: █
```

**Thanks everyone for
visiting, please support us
by clicking the follow
button, and please give
feedback(❤️) on this
document if you like it.**